WEEK-8

Agenda:
1. MGF
2. Distribution of sum of rand. var.
3. CLT
4. Z-score and Normal approximation
5. Gamma, Cauchy, Beta,
6. LC. of rand. var.

Moment Generating Functions (MGF)

Let X be a zero-mean random variable. The MGF of X, denoted $M_X(\lambda)$, is a func. from R to R defined as;

$$\boxed{M_X(\lambda) = E[e^{\lambda x}]}$$

- X: Discrete with PMF f_X
  $\Rightarrow$ X takes values $\{x_1, x_2, \ldots\}$
  $$M_X(\lambda) = f_X(x_1) e^{\lambda x_1} + f_X(x_2) e^{\lambda x_2} + \ldots \approx \sum P(X_i) \cdot e^{\lambda x_i}$$
- X: Continuous with PDF f_X and support T_X
  $$M_X(\lambda) = \int_{x \in T_X} f_X(x) \cdot e^{\lambda x} dx$$

Q.(1) Let $X \sim \text{Uniform}(\{1,2,3\})$, Find the MGF of centralised X.

Sol. $X \sim \{1,2,3\}$, $\boxed{Y = X - E(X)}$

$P(X=1) = P(X=2) = P(X=3) = \frac{1}{3}$

$$E(X) = \mu = \frac{1+2+3}{3} = \frac{6}{3} = 2$$

So, $\boxed{Y = X - 2}$

X	Y = X − 2	P
1	−1	1/3
2	0	1/3
3	1	1/3

(3) $M_Y(\lambda) = E[e^{\lambda Y}]$

(4) Compute expectation:

$$M_Y(\lambda) = \frac{1}{3}\left(e^{\lambda(-1)} + e^{\lambda(0)} + e^{\lambda(1)}\right) = \frac{1}{3}(e^{-\lambda} + 1 + e^{\lambda})$$

Q(2). Let $x_1, x_2, x_3 \sim$ i.i.d X and $X \sim$ Uniform$(\{-0.5, 0.5\})$,
Define $S = X_1 + X_2 + X_3$, find the MGF of S.

Sol.

$X_1 \simeq X_2 \simeq X_3$

$\boxed{M_X(\lambda) = E(e^{\lambda X})}$

$$M_X(\lambda) = P(X=-0.5) \cdot e^{\lambda(-0.5)} + P(X=0.5) \cdot e^{\lambda(0.5)}$$

$$= \boxed{\frac{1}{2}\left[e^{-0.5\lambda} + e^{0.5\lambda}\right]}$$

[i.i.d principle] ; $M_S(\lambda) = M_{X_1}(\lambda) \cdot M_{X_2}(\lambda) \cdot M_{X_3}(\lambda) = [M_X(\lambda)]^3$

$$= \left[\frac{1}{2}\left[e^{-0.5\lambda} + e^{0.5\lambda}\right]\right]^3$$

Q(3). Let $X_1, X_2, X_3 \sim$ i.i.d X and $X \sim$ Uniform$(\{-1, 1\})$.
Define $Y = X_1 + X_2 + X_3$. find the distribution of Y.

Sol.

$M_Y(\lambda) = [M_X(\lambda)]^3$ — (i)

$M_X(\lambda) = E(e^{\lambda x})$

$$M_X(\lambda) = P(X=-1) \cdot e^{\lambda(-1)} + P(X=1) \cdot e^{\lambda(1)}$$

$$= \frac{1}{2} \cdot e^{-\lambda} + \frac{1}{2} e^{\lambda} = \frac{e^{-\lambda} + e^{\lambda}}{2} \quad \text{— (ii)}$$

Put (ii) in (i)

(i.i.d) $\quad M_Y(\lambda) = \left[\frac{e^{-\lambda} + e^{\lambda}}{2}\right]^3$

$$\Rightarrow \frac{1}{8}\left[e^{-3\lambda} + e^{3\lambda} + 3e^{-2\lambda} \cdot e^{\lambda} + 3e^{-\lambda}(e^{\lambda})^2\right]$$

$$\Rightarrow \left(\frac{1}{8}\right)(e^{-3\lambda}) + \left(\frac{1}{8}\right)e^{3\lambda} + \frac{3}{8}e^{-\lambda} + \frac{3}{8}e^{\lambda}$$

$\Rightarrow P(X=-3) \cdot e^{\lambda(x=-3)} + P(X=3) \cdot e^{\lambda(x=3)} + P(X=-1) \cdot e^{\lambda(x=-1)} + P(X=1) \cdot e^{\lambda(x=1)}$

$X=-3 \Rightarrow \frac{1}{8}$ $\quad$ $X=-1 \Rightarrow \frac{3}{8}$

$X=3 \Rightarrow \frac{1}{8}$ $\quad$ $X=1 \Rightarrow \frac{3}{8}$

Ans $\Rightarrow$

Y	-3	-1	1	3
$P(Y=y)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

What is the value of 6th moment of the Normal(0, 3)?

Sol. $\mu = 0 \quad \sigma^2 = 3 \quad \sigma = \sqrt{3}$

$$E[X^n] = \frac{(\sigma)^n \cdot n!}{(2)^{n/2} \cdot \left(\frac{n}{2}\right)!}$$

$$\Rightarrow \frac{(\sqrt{3})^6 \cdot 6!}{(2)^{6/2} \left(\frac{6}{2}\right)!} \Rightarrow \frac{(27)(720)}{(8)\cdot(6)} \Rightarrow \boxed{405}$$

MGF of Sample Mean

Samples: $X_1, \dots, X_n \sim$ i.i.d X, $M_X(\lambda) = \frac{e^{\lambda/2} + e^{-\lambda/2}}{2}$

$\Rightarrow$ Sample mean: $\bar{X} = \frac{(X_1 + X_2 \dots + X_n)}{n}$

$$\Rightarrow M_{X/n}(\lambda) = \frac{e^{\lambda/2n} + e^{-\lambda/2n}}{2}$$

- $M_{\bar{X}}(\lambda) \to 1$ as n increases
- MGF converges at $\frac{1}{\sqrt{n}}$ scaling

$$\boxed{M_{\bar{X}}(\lambda) = \left(\frac{e^{\frac{\lambda}{2n}} + e^{\frac{-\lambda}{2n}}}{2}\right)^n}$$

Central Limit Theorem MGF

Let $X_1, \dots, X_n \sim$ i.i.d X with $E[X] = 0$, $Var(X) = \sigma^2$

Let $Y = (X_1 + \dots + X_n)/\sqrt{n}$. Then,

$$\boxed{M_Y(\lambda) \to e^{\lambda^2 \sigma^2/2}}$$

- MGF of $N(0, \sigma^2)$: $e^{\lambda^2\sigma^2/2}$
- Y is said to converge in distribution to Normal$(0, \sigma^2)$

Q. Use the following standarard Normal distributions

$\hookrightarrow F_Z(0.2617) = 0.603$
$F_Z(1.6) = 0.9452$
$F_Z(1.5) = 0.933$

The avg. life of a coffee machine is 5 yrs, with a S.D of 1 year. Assuming that the lives of these machines follow approximately a normal distribution, find the probability that the mean life of a random sample of 16 such machines falls b/w 4.6 and 5.4 years.

Sol. $\mu_{\bar{x}} = 5$ $\qquad P(4.6 \le \bar{x} \le 5.4) = ?$

$\sigma = 1$ $\qquad S.E_{\bar{x}} \Rightarrow \frac{\sigma}{\sqrt{n}} \Rightarrow \frac{1}{\sqrt{16}} = \frac{1}{4} = 0.25$

$n = 16$ $\qquad \uparrow$ Standard error

$x_1 = 4.6$	$z_1 = -1.6$
$x_2 = 5.4$	$z_2 = +1.6$

$$z_1 = \frac{x - \mu_{\bar{x}}}{SE_{\bar{x}}} = \frac{4.6 - 5}{0.25} \Rightarrow \boxed{-1.6}$$

$$z_2 = \frac{x - \mu_{\bar{x}}}{SE_{\bar{x}}} = \frac{5.4 - 5}{0.25} = \boxed{+1.6}$$

$$P(4.6 \le \bar{x} \le 5.4) = P(z \le 1.6) - P(z \le -1.6)$$
$$\Rightarrow P(z \le 1.6) - [1 - P(z \le 1.6)]$$
$$\Rightarrow 2 \cdot P(z \le 1.6) - 1$$
$$\Rightarrow 2 \cdot (0.9452) - 1 = 0.8904 \approx 0.89$$

Q. Let $x_1, x_2, \ldots, x_{50} \sim$ i.i.d. Poisson (0.04) and let $y = \sum_{i=1}^{50} x_i$. Use CLT to find $P(y > 8)$.

Sol.

$\lambda = 0.04$
$\mu = \lambda$
$\sigma^2 = \lambda$

$\mu = 0.04 \quad \sigma^2 = 0.04$

$\mu_y = 0.04 \times 50 = 2$

$\sigma_y^2 = 50 \times 0.04 = 2 \Rightarrow \sigma_y = \sqrt{2}$

② Apply CLT: $z = \frac{y - \mu}{\sigma} = \frac{8 - 2}{\sqrt{2}} = 4.24$

③ $P(y > 8) = P(z > 4.24)$
≈ 0
from z-table

Ans. 0.00

Q.(2) The random variable X representing the no. of cherries in a cherry puff has the following probability distribution:

x	4	6	8	10
$P(X=x)$	0.3	0.1	0.4	0.2

find the probability that the avg. no. of cherries in 64 cherry puffs will be less than 7.

Sol. $P(\bar{X} < 7)$ $\quad \bar{\mu}_x = 7$

$n = 64$

$$\mu_{\bar{x}} = E(x) \Rightarrow x_1 \cdot P(x=x_1) + x_2 \cdot P(x=x_2) + x_3 P(x=x_3) + x_4 \cdot P(x=x_4)$$

$$\Rightarrow 4(0.3) + 6(0.1) + 8(0.4) + 10(0.2) \Rightarrow 7$$

$\sigma_x = S.D$

$$(\sigma_x)^2 = E[x^2] - [E[x]]^2$$

var

$\Rightarrow 4.6$

$$\Rightarrow \boxed{\sigma_x = \sqrt{4.6} = 2.14}$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$\boxed{\frac{2.14}{\sqrt{64}} \Rightarrow 0.2675}$$

$$Z = \frac{\bar{X} - \mu_x}{SE_{\bar{x}}} = \frac{7-7}{0.275} = 0$$

$P(\bar{X} < 7) \Leftrightarrow P(z \leq 0) \rightarrow \boxed{0.5}$ → from z-table

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Q.(3) An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with mean equal to 600 hrs and a S.D of 50 hrs; find the probability that a random sample of 25 bulbs will have an avg. life of less than 615 hrs.

Sol. $\mu_{\bar{x}} = 600$ $\quad \sigma_x = 50$ $\quad n = 25$ $\quad P(\bar{X} < 615) = ?$

$$z = \left(\frac{\bar{x} - \mu}{SE_{\bar{x}}}\right) = \frac{615 - 600}{10} = 1.5$$

$$\left\{ SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{50}{\sqrt{25}} = 10 \right.$$

$P(z \leq 1.5) \Rightarrow 0.9332 \simeq 0.93$

Q.(4). The no. of accidents in a certain city is modeled by a Poisson random variable with a avg. rate of 9 accidents per day. Suppose that the no. of accidents on different days are independent. Use the CLT to find the probability that there will be more than 3300 accident in a certain year. Assume that there are 365 days in a year.

Sol. $\mu_{\bar{X}} \Rightarrow 365 \times 9 = 3285$

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{SE_{\bar{X}}} \Rightarrow \frac{3300 - 3285}{57.3} \Rightarrow \frac{15}{57.3} \simeq 0.261$$

$P(X > 3300)$

$\Downarrow$

$P(Z > 0.261)$

$\Rightarrow 1 - [Z \leq 0.261]$

$\Rightarrow 1 - 0.602 \Rightarrow 0.398 \approx \boxed{0.39}$

Linear combination of independent samples/Normals

$X_1, \ldots, X_n \sim$ ind. Normal

Mean: $E[Y] = \sum a_i \mu_i$

$Var(Y) = \sum a_i^2 \sigma_i^2$

- let $X_i \sim Normal(\mu_i, \sigma_i^2)$
- Suppose $\boxed{Y = a_1 X_1 + \cdots + a_n X_n}$ → also normal

→ Linear combination of ind. normals.

- Then,

$$Y \sim Normal(\mu, \sigma^2),$$

where $\mu = E[Y] = a_1 \mu_1 + \cdots + a_n \mu_n$

$\sigma^2 = a_1^2 \sigma_1^2 + \cdots + a_n^2 \sigma_n^2$.

- Linear combination of ind. normals is normal

Gamma Distribution (Y)

$X \sim Gamma(\alpha, \beta)$

- PDF → $f(x) \propto x^{\alpha - 1} e^{-\beta x}$, $x \geq 0$
- Parameters : $\alpha \to$ shape, $\beta \to$ rate
- ***Key Properteis:***
  - Mean $= \alpha / \beta$
  - Var $= \alpha / \beta^2$
- Imp Results:
  - ~~Sum of exponentials → Gamma~~
  - ~~Sum of Gamma → Gamma~~
  - Sum of n iid $Exp(\beta)$ is Gamma(n, β) → Gamma$(n\alpha, \beta)$
  - Square of Normal$(0, \sigma^2)$ is Gamma$(1/2, 1/2\sigma^2)$

Cauchy Distribution: → $X \sim \text{Cauchy}(\theta, \alpha^2)$

- PDF
$$f(x) = \frac{\alpha}{\pi\left(\alpha^2 + (x-\theta)^2\right)}$$

- Special properties:
  - Mean × undefined
  - Var × undefined
  - Mgf × does not exist

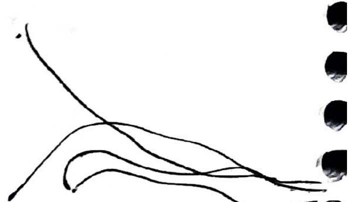
Beta Distribution

- Domain : $0 < x < 1$
- PDF : $f(x) \propto x^{\alpha-1}(1-x)^{\beta-1}$
- μ and var :
  - Mean $= \alpha/(\alpha+\beta)$
  - Var $= \alpha\beta / \left[(\alpha+\beta)^2(\alpha+\beta+1)\right]$
- Key identity: if :
  - $X \sim \text{Gamma}(\alpha)$
  - $Y \sim \text{Gamma}(\beta)$

  Then: $\frac{X}{X+Y} \sim \text{Beta}(\alpha, \beta)$

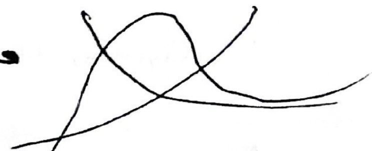
Distribution shapes

Normal → Bell cone
Cauchy → heavy tails
Gamma → skewed right
Beta → flexible




Beta


→ Normal



Sample mean and var. of Normal Samples

Suppose $X_1, \ldots, X_n \sim \text{Normal}(\mu, \sigma^2)$. Then,

1. $\bar{X} \sim \text{Normal}(\mu, \sigma^2/n)$.
2. $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$, Chi - square with n-1 degree of freedom
3. $\bar{X}$ and S^2 are independent.

Q. Let X_1, X_2, X_3, X_4 and $X_5 \sim$ iid Gamma$(2,5)$. The mean of Gamma(α,β) is $\frac{\alpha}{\beta}$. Find the mean of the distribution $X_1+X_2+X_3+X_4+X_5$.

Sol. $X_1, X_2, X_3, X_4, X_5 \sim$ i.i.d Gamma $(2,5)$

$X_1+X_2+X_3+X_4+X_5 =$ Gamma $(n\alpha, \beta)$

$=$ Gamma $(5(2), 5)$

$=$ Gamma$(10,5)$

mean $= \alpha/\beta = 10/5 = 2$

Q. Let $X_1, X_2, X_3 \sim$ iid Normal $(0,9)$. Find the mean of $X_1^2+X_2^2+X_3^2$

Sol. $X \rightarrow$ Normal $(0, 9)$; μ, σ^2

$X^2 \rightarrow$ Gamma $\left(\frac{1}{2}, \frac{1}{2\sigma^2}\right)$

$X_1^2+X_2^2+X_3^2 \Rightarrow$ Gamma $\left(\frac{n}{2}, \frac{1}{2\sigma^2}\right)$

$\Rightarrow$ Gamma $\left(\frac{3}{2}, \frac{1}{2(9)}\right) \Rightarrow$ Gamma $\left(\frac{3}{2}, \frac{1}{18}\right)$

Ans. $\mu = \frac{\alpha}{\beta} = \frac{3/2}{1/18} = 27$

Q. Let $X_1, X_2, X_3 \cdots X_{500} \sim$ i.i.d Normal $(0,1)$.
Evaluate $P(X_1^2+X_2^2+X_3^2+\cdots+X_{500}^2 > 520)$ using CLT.
Hint: $(X_1^2+X_2^2+X_3^2+\cdots+X_{500}^2) \sim$ Gamma $(250, 0.5)$.

Sol. $S \sim$ Gamma (α, β) ; $\alpha \to 250$, $\beta \to 0.5$

$\mu = \frac{250}{0.5} = 500$; α/β

$\sigma^2 = \frac{250}{(0.5)^2}$; α/β^2 $= 31.62^2$

$\sigma = 31.62$

Apply CLT:

We approximate: $S \approx N(500, 1000)$; μ, σ^2

Standardize: $Z = \frac{520-500}{31.62} \Rightarrow 2.21$

$P(S > 520) = P(Z > 2.21) \approx 0.0136$

$\boxed{0.01}$

Questions

MGF

Graded Q. 10
(1)

(MGF Transformation Rule)

Q. Let the MGF of X be given as $M_X(\lambda) = e^{5\lambda + 7\lambda^2}$. Define a new random variable $Z = 3X + C$. Which of the following are true for the MGF of Z, $M_Z(\lambda)$?

Sol. $\boxed{\text{MGF Tranformation Rule}}$

$$Z = \overset{3}{a}X + \overset{c}{b}$$

Formula: $M_Z(\lambda) = e^{b\lambda} \cdot M_X(a\lambda)$

$$M_Z(\lambda) = e^{c\lambda} \cdot M_X(3\lambda)$$

$$M_X(3\lambda) = e^{15\lambda + 63\lambda^2}$$

$$\boxed{M_Z(\lambda) = e^{c\lambda + 15\lambda + 63\lambda^2}}$$

Graded 11.
Q.

x's PMF ⇒

x	0	2	3
$f_X(x)$	1/5	3/5	1/5

Suppose $X_1, X_2 \sim$ i.i.d. X.
Define a random var. $Y = X_1 + X_2$

✓ $M_Y(\lambda) = M_{X_1}(\lambda) \times M_{X_2}(\lambda)$

✓ $M_Y(\lambda) = (M_X(\lambda))^2$

⇒ Compute probabilities (convolution)

- $P(Y=0) = \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$
- $P(Y=2) = 2 \cdot \frac{1}{5} \cdot \frac{3}{5} = \frac{6}{25}$
- $P(Y=3) = 2 \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{2}{25}$
- $P(Y=4) = \frac{3}{5} \cdot \frac{3}{5} = \frac{9}{25}$
- $P(Y=5) = 2 \cdot \frac{3}{5} \cdot \frac{1}{5} = \frac{6}{25}$
- $P(Y=6) = \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$

$M_Y(\lambda) = \Sigma P(Y=y)\, e^{\lambda y}$

$$\Rightarrow \frac{1}{25} + \frac{6}{25}e^{2\lambda} + \frac{2}{25}e^{3\lambda} + \frac{9}{25}e^{4\lambda} + \frac{6}{25}e^{5\lambda} + \frac{1}{25}e^{6\lambda}$$

CLT

Graded

Q.7. Let the time taken (in hrs) for failure of an electic bulb follow the exponential distribution with the parameter 0.05. Suppose that 200 such light bulbs say $L_1, L_2 \dots L_{200}$ are used in the following manner. For every i, as soon as the light L_i fails, L_{i+1} become operative, where $i: 1 \to 199$ [i.e. If L_1 fails, L_2 becomes operative, if L_2 fails, L_3 becomes operative, and so on]. Let the total time of operation of 200 bulbs be denoted by T. Using CLT compute the probability that T exceeds 1600 hrs.

Sol. $L_i \sim \text{Exponential}(0.05)$

$T = L_1 + L_2 + \dots + L_{200}$

Total Time T = sum of 200 i.i.d exponential rv.

$\mu = 1/\lambda = 20$

$\sigma^2 = 1/\lambda^2 = 400$

for sum T: ⇒ $200 \cdot 20 = 4000$

for σ_T: $\sqrt{200 \cdot 400} = \sqrt{80000} \approx 282.84$

↳ behaves like S.E in this ques

③ Apply CLT: $T \approx N(4000, 80000)$

$P(T > 1600)$

Standardize: $Z = \dfrac{1600 - 4000}{282.84} \approx -8.5$

Z-table

↳ row → first 2 digit

col → last decimal

$P(T > 1600) = P(Z > -8.5) = 1 - [P(Z \leq -8.5)]$

$P(Z > a) = 1 - F_Z(a) = \boxed{1 - F_Z(-8.5)}$

Ans.

see Q.9 for good z-table value getting

Gamma Q.5

$X \sim \text{Gamma}(\alpha = 2n, \beta = 1)$

$E[x] = \dfrac{\alpha}{\beta} = 2n \qquad Var(x) = \dfrac{\alpha}{\beta^2} = 2n$

$P\left(\left|\underbrace{\frac{X}{2n}}_{Y} - 1\right| > 0.01\right)$

$E[Y] = 1$

$Var[Y] = \dfrac{Var(X)}{(2n)^2} = \dfrac{1}{2n}$

$P(|Y-1| > 0.01) \leq \dfrac{Var(Y)}{(0.01)^2}$

$= \dfrac{1/(2n)}{0.0001} = \dfrac{1}{0.0002n} < 0.01$

↳ $n > 500000$